

A Survey on Retrieval-Augmented Generation in Applications of Education and Teaching

Feng Pan¹, Qiyun Zhou², Weitong Guo^{1,3*}, Hongwu Yang^{1,3}

¹School of Educational Technology, Northwest Normal University, Lanzhou 730070, China

²Faculty of Artificial Intelligence in Education, Central China Normal University, Wuhan 430000, China

³Key Laboratory of Education Digitalization of Gansu Province, Lanzhou 730070, China

Corresponding author: guowt@nwnu.edu.cn

Abstract—Retrieval-augmented generation (RAG) seamlessly combines information retrieval with generative artificial intelligence, allowing large language models (LLMs) to enhance the accuracy and relevance of responses by dynamically retrieving and integrating contextual knowledge from external databases. In education, RAG holds transformative potential, facilitating personalized learning path recommendations, automated question-answering systems, knowledge retrieval, generation of teaching resources, and adaptive evaluation frameworks. Its strengths, such as efficiency, precision, personalization, and interactivity, streamline the acquisition and utilization of educational resources, ultimately fostering improved pedagogical outcomes. Nonetheless, several challenges remain, such as integrating interdisciplinary knowledge, addressing generative hallucinations, and navigating ethical considerations in data usage. Future advancements are expected to concentrate on precision teaching support, adaptive learning ecosystems, data-driven educational governance, and specialized applications in areas like vocational training and inclusive education. RAG is positioned to reshape educational practices by addressing technical and pedagogical barriers and promoting equitable access and personalized learning experiences in an evolving digital landscape.

Index Terms—Intelligence education, Retrieval-augmented generation, Large language model, Educational application

I. INTRODUCTION

Education is experiencing a significant transformation driven by rapid advancements in artificial intelligence technology. Technological advancements have continually reshaped educational models, from early Computer-Assisted Instruction to the current focus on Generative Artificial Intelligence (GAI). Recently, Retrieval-Augmented Generation (RAG)-based technologies, with their unique “retrieval-generation” collaborative mechanism, have emerged as a leading area of research in educational intelligence.

Traditional educational technologies face two main challenges. First, static knowledge bases often fail to meet the dynamic needs of teaching scenarios, leading to a decline in response quality due to insufficient data coverage, especially in open-domain questions and interdisciplinary integration. Second, pure generative models’ black-box nature increases the risk of factual errors in the generated content [1], particularly in educational contexts that prioritize accuracy. RAG offers a novel approach to tackling key educational challenges

by dynamically integrating structured knowledge with generative model capabilities. It enhances knowledge accuracy by retrieving precise information from extensive resources, such as textbooks and academic papers, effectively addressing the “hallucination” problem typical of traditional generative models [2]. RAG also enables personalized adaptation by creating tailored learning content and guidance based on students’ progress and preferences. Additionally, RAG fosters real-time communication among students, teachers, and administrators, significantly enhancing feedback and interaction efficiency. Its knowledge verification-logical reasoning-content generation-based architecture provides unique advantages for intelligent teaching, adaptive learning, and resource management.

Given RAG’s educational potential, a comprehensive review of its applications is crucial. This study employs a literature review method to analyze RAG’s educational research from 2024-2025, assessing strengths, challenges, and future research directions to inform educators and researchers about trends and innovations in RAG technology in education.

II. OVERVIEW OF LLMs AND RAG

A. Overview of the LLMs

Large Language Models (LLMs) are deep learning-based models with large parameter scale, relying on massive training data, capable of capturing complex linguistic laws and realizing knowledge representation, with text generation, question and answer, etc., presenting the characteristics of large parameter scale, data diversity, strong generalization ability and relying on high-performance computing.

Its development can be divided into five stages, as outlined in Table I: the early exploration period before 2017, the introduction of the Transformer architecture is a key breakthrough [3], which solves the limitations of RNN, LSTM and other models to deal with long text; the pre-training paradigm rises in 2018-2019, and the pre-training + fine-tuning models of GPT-1 and BERT mode of GPT-1 and BERT improves the performance of NLP tasks [4]; 2020-2022 model parameter explosion growth, GPT-3 and other models show emergence ability, parameter scale expansion significantly improves the task generalization and adaptability, and during the same period of time the emergence of Google T5, Microsoft Turing-NLG, ERNIE and other models. NLG, Microsoft Turing-NLG, ERNIE 3.0, Wudao 2.0 and other models have broken through

The research is supported by the research fund from the National Natural Science Foundation of China (Grant No. 62267008, No. 62067008).

the performance bottleneck of traditional models by expanding the parameter scale, and demonstrated stronger contextual understanding and generation capabilities; from 2023, the models will turn to multimodal fusion and vertical domain applications, while focusing on reducing training costs and ethical issues.

Currently, LLMs face challenges such as high computational cost, ethical risks, and environmental impact. Future development may focus on efficient training techniques, controlled generation, embodied intelligence, and accessibility enhancement. In essence, LLMs promote the transformation of AI from specialized narrow models to general intelligence and reshape human-computer interaction. With technological advances and improved governance, they are expected to unleash their potential in more fields and accelerate the process of social digitization.

TABLE I
THE DEVELOPMENT HISTORY OF LARGE LANGUAGE MODELS

Time Period	Stage Name	Key Developments & Representative Models
Before 2017	Early Exploration & Transformer Revolution	Dominance of RNN, LSTM, etc
2017-2018	Emergence of Transformer	Google proposed the Transformer architecture.
2018-2019	Rise of Pre-training Paradigms	GPT-1 and BERT introduced.
2020-2022	Explosion of Parameter Scale	GPT-3, T5, Turing-NLG, ERNIE3.0, WuDao 2.0, etc.
2023-Present	Multimodal & Vertical Development	Breaking through single-text modality, moving toward multimodal integration.

B. Overview of Retrieval Augmented Generation

In the field of natural language processing, RAG realizes technological breakthroughs by integrating large language models with efficient retrieval systems. Compared with traditional models relying on static training data, its advantage lies in the real-time integration of dynamic external information when generating content, which significantly improves the accuracy, relevance and contextualization of the output. As illustrated in Fig. 1, The RAG architecture consists of three core modules: a knowledge base that provides data support, an extractor that is responsible for information retrieval, and a generator that synthesizes responses based on a large language model. The workflow is as follows: when the system processes a user query, the extractor synchronously scans the knowledge base to extract relevant information, fuses it with the original query to form a contextual context, and the generator generates a structured response based on the augmented context and provides feedback to the user. In terms of technology evolution, RAG has iterated from the basic form to the modular stage, a process that reflects both architectural optimization and application scenario expansion and performance improvement.

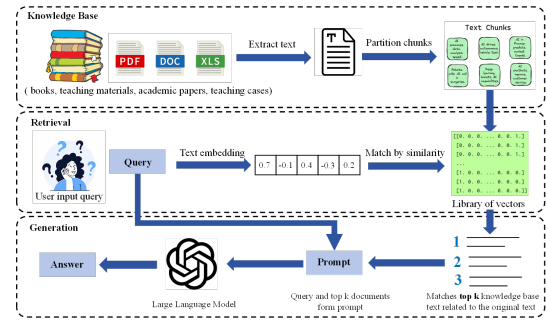


Fig. 1. The principle of Retrieval Augmented Generation.

III. CURRENT STATUS OF RAG APPLICATIONS IN EDUCATION AND TEACHING

By integrating LLMs with external knowledge retrieval, RAG enhances educational technology tools' accuracy, relevance, and personalization. This paper reviews the advancements in RAG within the educational sector based on recent studies, focusing on four key aspects: personalized learning path recommendations, automated question-answering and knowledge retrieval, automated teaching resource generation, and automated evaluation and feedback processes.

A. Personalized Learning Path Recommendation

Students exhibit various learning backgrounds and abilities, making the traditional one-size-fits-all teaching model inadequate. RAG effectively addresses this challenge by gathering and integrating learner profiles, progress, and interests to understand each student's unique context. It then accurately fetches and matches relevant learning resources, creating personalized learning pathways. For instance, [5] utilized the RAG model to efficiently extract information from a university's internal datasets, including academic papers, course materials, and student profiles. Similarly, [6] applied RAG to assist high school students with college application planning, tailoring pathways to their specific circumstances and broadening RAG's application within education. Moreover, [7] introduced SKYRAG, which combines an advanced retrieval mechanism with LLMs to fetch course materials from MOOC platforms and align them with learner profiles, further personalizing educational paths. Additionally, [8] integrated RAG with customized LLMs to develop an intelligent chatbot. By leveraging LlamaIndex to retrieve academic information, they optimized responses and enhanced interactivity, resulting in a more efficient personalized learning experience. Personalized learning path recommendations derived from RAG provide tailored suggestions and eliminate redundant and irrelevant content, improving overall learning efficiency. This approach fosters learners' enthusiasm and engagement, enhances the educational experience, supports individuals in achieving their academic goals, promotes personalized education development, and ultimately boosts overall learning outcomes.

B. Automatic Question-Answering and Knowledge Retrieval

In automatic question-answering and knowledge retrieval, RAG first enables the model to comprehend user questions effectively. It then retrieves information from various sources, including course materials and online texts, generating accurate and contextually relevant answers. For instance, [9] developed an AI intelligent assistant using the RAG model to address student inquiries. This assistant provides precise, context-aware suggestions by synthesizing information from school documents with the specific context of each student's question. In a related study, [10] explored the application of the RAG model in the role of virtual teaching assistants, laying the groundwork for its use in personalized education. Following that, [11] designed RAG-based tutoring software for statistics aimed at helping students grasp quantitative analysis methods effectively. Similarly, [12] applied RAG in K-12 education, utilizing course materials as data sources. Their research demonstrated that RAG could facilitate intelligent tutoring by delivering timely and relevant information, thus expanding its applicability in basic education. The RAG model's application in automatic question-answering and knowledge retrieval offers efficient and precise support within the educational sector. It minimizes the "hallucinations" associated with large language models, enhances answers' accuracy and relevance, addresses information obsolescence issues, and broadens access to knowledge. This significantly optimizes the overall learning and information retrieval experience.

C. Automated Teaching Resource Generation

Applying the RAG in automated teaching resource generation allows for extracting pertinent information from extensive knowledge bases, facilitating the automatic creation of high-quality teaching content through integrating retrieval and generation technologies. For instance, [13] employed a localized knowledge base within the RAG framework to establish query contexts for prompts directed at large language models (LLMs). This study illustrates that RAG offers substantial advantages in generating automated teaching resources and providing valuable support for educators. Traditional education and many intelligent tutoring systems often emphasize the one-way transmission of knowledge, which can neglect the development of student's independent learning skills. To combat this limitation, [14] utilized RAG to generate heuristic questions based on retrieved knowledge paths, encouraging interactive thinking and enhancing students' autonomous learning capabilities. Additionally, the manual construction of questions is typically complex and time-consuming. In response, [15] integrated RAG with LLMs to create a prompt pattern derived from a collective knowledge base shared by teachers. This approach exemplifies how the RAG model, through accurate case recommendations, generating heuristic question sequences, and constructing high-quality questions, can significantly enhance the efficiency and quality of teaching resource production. Overall, the application of RAG streamlines the automated generation of teaching materials and fosters the development of students' independent learning

abilities, providing robust technical support for the educational sector.

D. Automated teaching evaluation and feedback

Evaluating teachers' instruction and students' learning is essential in educational settings. The RAG model efficiently retrieves information from extensive educational resource repositories and automatically conducts comparative performance analyses, including submitted student assignments and recorded teaching videos. Subsequently, it generates assessment reports and feedback, highlighting strengths, weaknesses, and areas for improvement, facilitating the growth of both teachers and students. For example, [16] integrated RAG into an LLM while teaching the Unified Modeling Language course to enhance students' modeling skills and behaviors. Building on this, [17] explored using RAG combined with fine-tuning techniques for automated test correction, enabling the automated scoring of subjective assessments. Additionally, [18] employed RAG and LLMs to evaluate student performance and provide targeted feedback, resulting in more accurate and comprehensive evaluations. Addressing the limitations of traditional simulated teaching evaluations, [19] proposed a RAG framework combined with a local LLM to analyze the frequency of simulated teaching practices. This research highlights a promising method to enhance educational assessment through advanced language technology. The application of RAG in educational assessment and feedback is continually evolving. From intelligent tutoring to automated scoring, comprehensive evaluation, and simulated teaching analysis, RAG demonstrates significant potential for improving educational quality and efficiency.

IV. ADVANTAGES AND CHALLENGES OF RAG IN EDUCATION AND TEACHING

RAG holds significant promise for enhancing educational practices, driving innovation in teaching models, and improving the learning experience. However, it also faces several challenges that must be addressed.

A. Personalized Learning Path Recommendation

Personalized learning path recommendations are highly adaptive, utilizing students' learning data and interests to customize their educational journeys. For instance, the system might suggest courses focused on classic literature appreciation and writing for literature enthusiasts to enhance engagement and motivation. RAG excels in dynamic adjustment by refining learning paths in real-time based on students' progress. If a student masters a chapter ahead of schedule, the system may advance to more challenging content; conversely, if a student struggles with foundational concepts, it will provide supplementary resources to ensure learning continuity. However, as extensive data collection could jeopardize student confidentiality, any data breach could lead to a trust crisis, making it essential to implement robust safeguards. Additionally, algorithmic bias remains a pressing issue. Overreliance on historical grades may overlook a student's potential, resulting

in misjudgments and recommending unsuitable learning paths that could hinder their growth.

B. Automatic question-answering and knowledge retrieval

RAG's key advantage is effectively transcending the limitations of traditional textbooks in automatic question-answering and knowledge retrieval by integrating multi-source knowledge. For example, students studying history or science can access diverse materials that enrich their understanding and foster knowledge expansion. RAG facilitates learning by allowing students to obtain answers anytime and anywhere, creating a seamless transition between classroom learning and independent study to help satisfy curiosity and significantly reduce the time spent searching for information. However, ambiguous or interdisciplinary questions can lead to misleading responses, confusing students and hindering their knowledge construction. Additionally, information overload is a concern; unfiltered retrieval results can overwhelm younger students, making it challenging to discern key information and impeding their ability to learn autonomously.

C. Automated teaching resource generation

Automated teaching resource generation can efficiently create materials such as courseware and exercises tailored to teachers' needs. For instance, teachers can use it to quickly produce batch lesson plans that align with the syllabus, significantly reducing educators' preparatory workload. Furthermore, incorporating advanced technologies can generate innovative resources like virtual reality scenarios and interactive animations, which enliven the classroom, capture students' attention, and enhance knowledge retention. However, challenges persist, particularly regarding copyright issues and content quality. Using diverse material sources may lead to copyright disputes, making the legal parameters for usage ambiguous. Automatically generated content may contain logical inconsistencies or factual inaccuracies, necessitating a thorough review by teachers to maintain the quality of instruction.

D. Automated teaching evaluation and feedback

RAG provides numerous benefits in automated teaching evaluation and feedback by efficiently processing large volumes of student learning data for automatic assessment, eliminating the tediousness and time-consuming nature of manual grading while significantly enhancing overall efficiency. This lets teachers quickly understand students' learning statuses, facilitating more informed instructional decisions. Moreover, leveraging its retrieval and analytical capabilities, RAG accurately references educational databases for multi-dimensional performance analyses by delivering comprehensive assessments and identifying specific issues to establish an objective foundation for teaching. RAG customizes feedback based on individual student data, recommends appropriate learning resources, and accommodates diverse learning paces and needs, fostering personalized growth. However, challenges persist, particularly regarding data quality and balancing human and machine responsibilities. The accuracy and integrity of input

data must be rigorously maintained; establishing robust data cleaning and review mechanisms is essential to prevent incorrect or missing data from distorting results. Continuous expansion and updating of the teaching resource database are also necessary to ensure access to the latest knowledge. Clarifying the boundaries of responsibilities between teachers and RAG is also crucial. Teachers should review the results and provide professional insight on complex issues, while RAG assists with basic assessments, maximizing their combined strengths for a more effective educational model.

V. FUTURE RESEARCH DIRECTIONS

Future research on RAG in education can focus on precision teaching support systems, adaptive learning Ecosystems, decision-making hubs for educational governance, educational research innovation platforms, special education empowerment systems, and mechanisms for advancing equity in education.

A. Precision Teaching Support System

Knowledge-Graph (KG)-Driven Question Answering: RAG can create a subject-specific KG database [20], extracting relevant knowledge through semantic matching and generating multi-modal analytical solutions. When a student poses a question, the system automatically retrieves information from the KG and associated knowledge base, forming a dynamic knowledge network that facilitates comprehensive answers.

Intelligent Reconstruction of Teaching Resources: Leveraging the course objectives teachers provide, RAG can automatically search through global open educational resources to generate customized courseware tailored to teaching needs to ensure the materials are aligned with the educational goals.

B. Adaptive Learning Ecosystem

Dynamic Modeling of Cognitive Trajectories: RAG constructs a personalized cognitive development map by continuously collecting and analyzing learning behavior data. It automatically recommends differentiated resources when it identifies a student experiencing a bottleneck in spatial imagination skills, particularly in solid geometry. These may include Augmented Reality (AR) applications for origami modeling and instructional videos that analyze architectural perspectives, effectively supporting students' learning needs.

Interdisciplinary Competence Cultivation Engine: RAG can dismantle subject knowledge barriers in project-based learning. For instance, exploring the "Silk Road" connects historical documents, geo-climatic data, trade models, and digital resources related to Dunhuang art, creating a comprehensive interdisciplinary learning pathway.

C. Decision-making hubs for educational governance

Policy Impact Simulation and Deduction: When a new education reform plan is introduced, RAG can search for global case studies of similar policies and generate a multi-dimensional impact prediction report using regional education data. This report encompasses indicators such as the necessary adjustments in teaching staff structure, the ratio of hardware investments, and changes in the education equity index.

Campus Risk Early-warning System: By analyzing educational administration data, public opinion insights, and security logs in real-time, RAG can predict potential risks, such as incidents of campus bullying and mental health crises.

D. Educational Research Innovation Platform

Tracking of Academic Frontiers: After researchers input preliminary hypotheses, RAG automatically searches cross-language literature and generates a research-status map.

Construction of Simulation Research Environment: Based on a massive educational case database, RAG can generate virtual experimental scenarios. Educational policy researchers can simulate the “Rural Teachers Rotation System” implementation effects in regions with different economic levels and obtain parameterized analysis reports.

E. Special Education Empowerment System

Multimodal Interaction Interface: For visually impaired learners’ sound-tactile feedback system, RAG can transform geometric figures into vibration frequency sequences, conveying spatial concepts via wearable devices.

Personalized Intervention Plan: For autistic children, RAG analyzes personal training videos to generate social-situation training scripts, automatically adjusts interaction difficulty, and pushes home-training guidelines.

F. Mechanisms for advancing equity in education

Dialect Intelligent Translation System: RAG creates a real-time translation feature using local dialect voice databases. It allows ethnic minority students to access quality education in their native tongues and concurrently generates bilingual learning materials.

Resource Gap Prediction Model: By analyzing regional education data, RAG forecasts subject-based teacher shortages, smartly recommends training for normal students, and produces course packs to boost teachers’ capabilities.

VI. CONCLUDE

This paper overviews the concepts and principles of LLMs and RAG for establishing a theoretical foundation for RAG’s educational applications and the challenges faced, including data privacy and security concerns, generated content accuracy and reliability, and adaptability to various educational contexts. Addressing these challenges will guide future research.

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